

Trained



train



Large labelled
training set

Environment-specific
features

A

Standard



Cross-env.



Training-free, prior work



Frozen patch
features

→



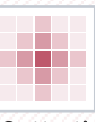
Codebook

or



Segmentation

or



CLS attention

Prior-selected / aggregated features

B

Standard



Cross-env.



Ours

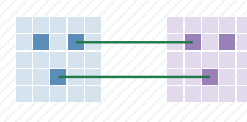


Frozen patch
features

→



Within-image
Balanced, matchable patch evidence



Across-image

C

Standard

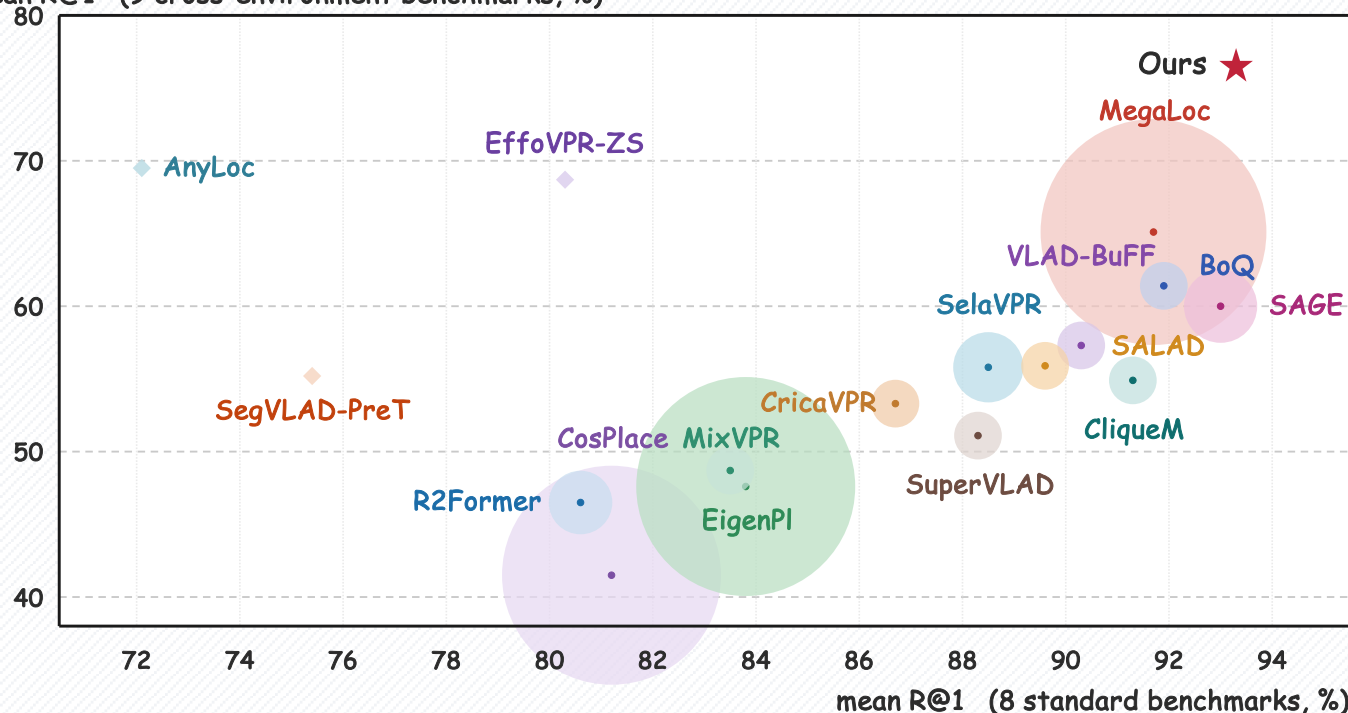


Cross-env.



Training cost vs. performance

mean R@1 (9 cross-environment benchmarks, %)



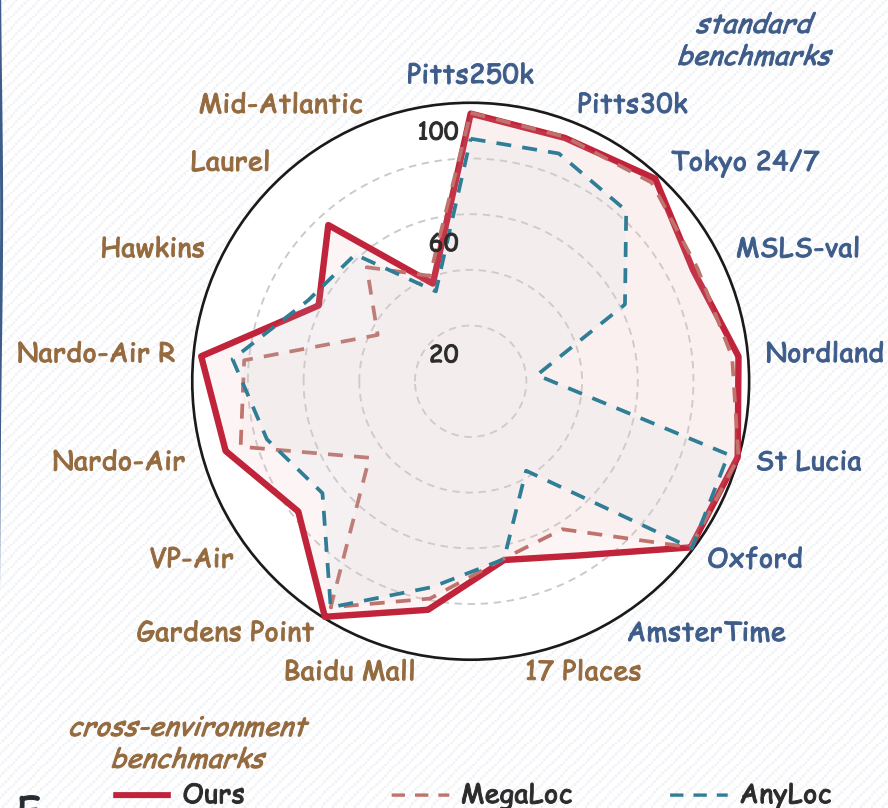
D

1 million images

5 million images

★ ours (training-free) ♦ training-free ● trained

Performance on 17 benchmarks



E

cross-environment
benchmarks

— Ours

- - - MegaLoc

- - - AnyLoc